

WHEN, WHY, AND HOW FAR IT HELPS

Emotion-Mediated Personalized Image Aesthetic Assessment

Pinwa



What we'll cover today.

1

The problem & our idea

Understanding individual taste disagreement and introducing the emotion mediation framework.

2

Does it work?

Evaluating the predictive gain of intermediate emotions and comparing with trait-based baselines.

3

When & why does it help?

Analyzing the driving mechanism, variance decomposition, and redundancy in stronger backbones.

4

How far can it go?

Estimating realistic cross-session noise ceilings and solving the cold-start rating threshold.

5

Is the gain real?

Verifying semantic validity using placebo control controls and explaining unusual users.

Taste is Personal.

STANDARD PARADIGM

IAA: Image Aesthetic Assessment

 Consensus Image

3.5 / 5.0

“One score for everyone”

PERSONALIZED PARADIGM

PIAA: Personalized Aesthetic Assessment

 Personalized Image

User A
5.0 / 5.0

User B
4.0 / 5.0

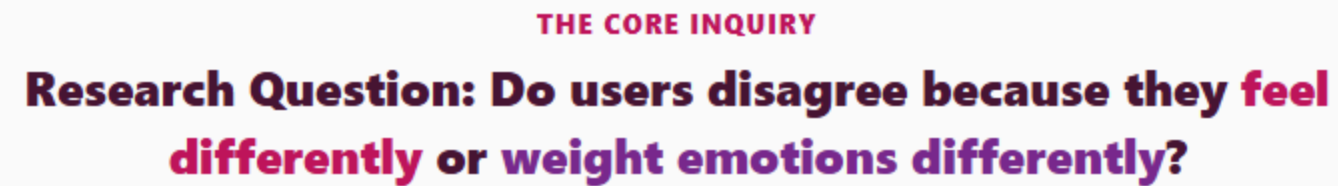
User C
2.0 / 5.0

“How much do YOU like it?”

Used in: Photo Applications • Recommendation Systems • Ranking AI-Generated Images



Emotion-Mediated PIAA | 4





shared (trained on general users)

• 07 — INTERPRETABILITY EXAMPLE

What does emotion explain?

• USER A (IMPRESSED)

The aesthetic preference is heavily driven by feeling **Impressed** (represented by the long pink positive bar).

• USER B (NOSTALGIC & AMUSED)

Aesthetic score is driven by **Nostalgic** and **Amused** (long purple positive bars).

“We can explain exactly WHY User A and User B rate the same image differently. A standard black-box model cannot reveal this.”

 Subjective Emotion Weights (User A vs User B)

POSITIVE BARS (RIGHT) PUSH BEAUTY UP • NEGATIVE BARS (LEFT) PULL IT DOWN

Because the last step is linear, we read each person's emotion weights

What one rating looks like

VISUAL INPUT (LANDSCAPE DOMAIN)

 Sample Landscape Image

Image ID: 3099 (XPASS-Vis Dataset)

USER EVALUATOR RESPONSE

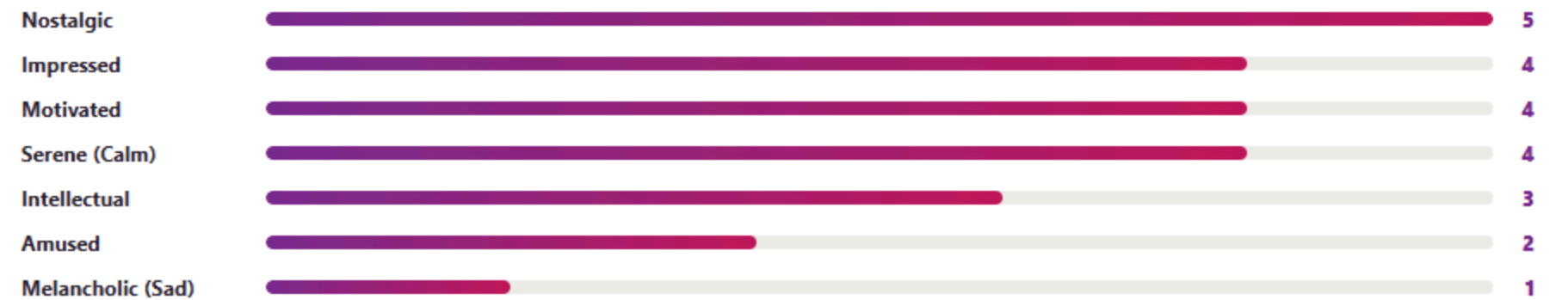
AESTHEMOS Framework (7 Core dimensions)

6 / 7

Aesthetic Score

Subjective beauty score rated on a 7-point Likert scale

7 Intermediate Emotions (Scale 1–5):



* each person gives a beauty score (1-7) and 7 emotion ratings (1-5) per image

Deep enough per person, and leak-free

129

People (Evaluators)

6,526

Images (Unique)

87,836

Ratings (Total)

3

Categories

Art / Fashion / Landscape

1

Deep Ratings Per Person

Each person rated 200+ images → deep enough for a personal formula.

2

Test-Retest Setup

Some images rated twice → used later to establish human noise ceilings.

PROTOCOL

leak-free: test users & images never seen in training

Part 2 — Does it actually work?



- 1 Does emotion help?
- 2 Comparable to baselines?
- 3 When does it help?
- 4 How far can it go?
- 5 Is the gain real?

- Perception vs. Weighting**
Understanding personal taste source
- Backbone Redundancy**
Evaluating larger vision LLMs
- Hard Users**
Analyzing high-variance profiles
- Trait-based Extensions**
Aesthetic-personality connections

Emotion-Mediated PIAA | 11

Going through emotion Helps 93% of the Time.

+0.066 **Significant Accuracy Gain**
Direct CCC (0.293) → Hybrid CCC (0.359)

Helped Share
92.8%
of user-domain units

Support Effect
Dose-Response
More ratings = higher gain

Dose-Response Effect: As user rating data grows, the accuracy improvement (Delta CCC) increases consistently. This confirms that collecting personalized preference data provides solid, scalable benefits.



Dose-Response Support Curve

Delta (Hybrid - Direct) increases with user ratings size

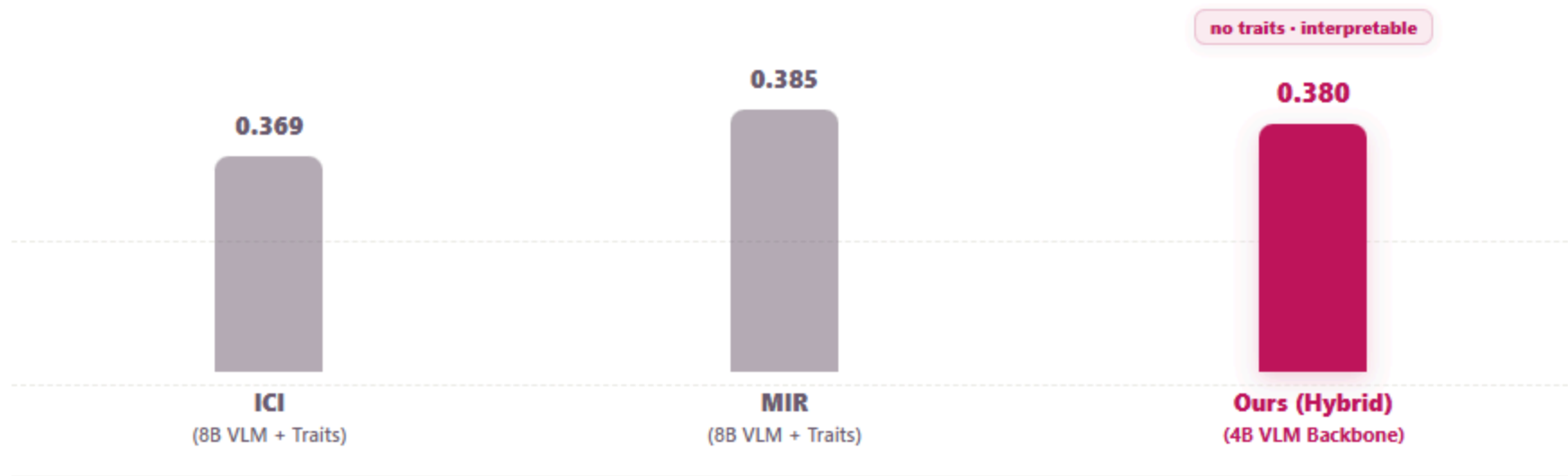
Direct CCC: 0.293

Hybrid CCC: 0.359

Helped Share: 92.8%

We match trait methods, Without Traits.

PERFORMANCE COMPARISON (CCC METRICS)



4B ties 8B — bigger backbone not needed

Core Novelty & Contribution

We achieve competitive predictive accuracy matching SOTA 8B baselines without requiring intrusive, multi-question user personality surveys (like the Big Five index).

STRONG ARGUMENT FOR USER PRIVACY

Protecting User Privacy

Predicting taste through intermediate aesthetic emotions allows us to completely bypass the storage or assessment of sensitive user profile traits.

It works. Now three questions.

- 1 When does it help?**
- 2 How far can it go?**
- 3 Is the gain real?**

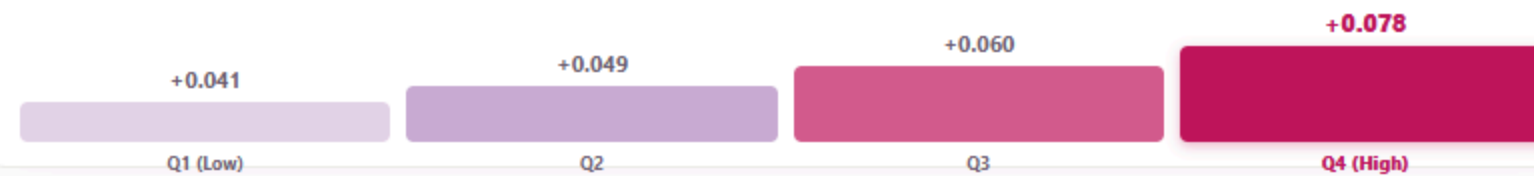
It helps more When We Read Emotions Well.

CORE DRIVING FACTOR

The more accurately the system predicts a specific user's emotions, the higher the aesthetic accuracy gain.

Spearman r
+0.273

EMOTION PREDICTABILITY (\$EMO_R\$) QUARTILE GAIN



one clean factor; the rival explanation was a confound



Gain vs Emotion Predictability (emo_r)

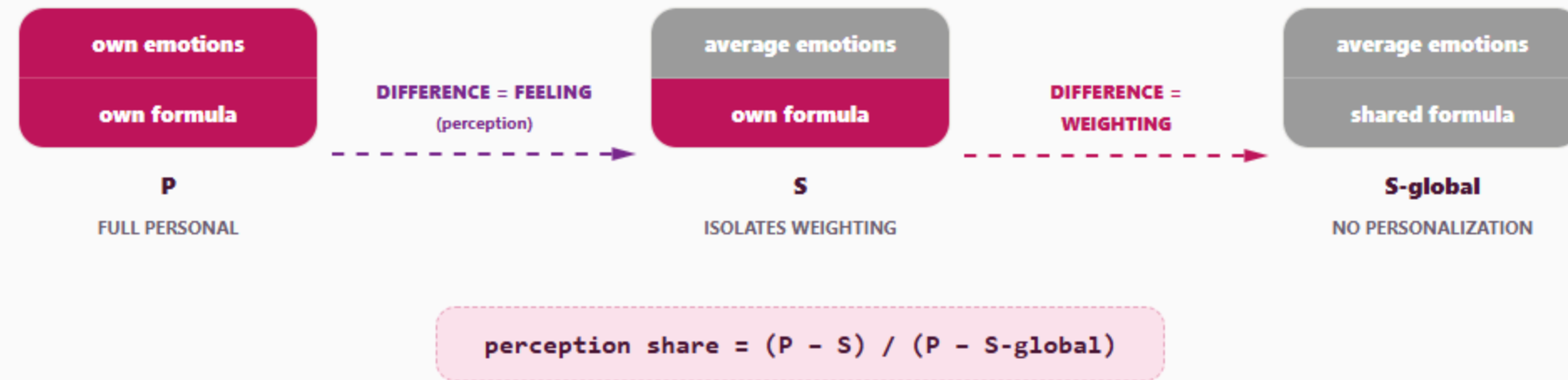
Delta (Hybrid - Direct) scales with emotion prediction accuracy (Spearman +0.273)

Spearman r: **+0.273**

Q1 Gain: **+0.041**

Q4 Gain: **+0.078**

Where does personal taste come from?

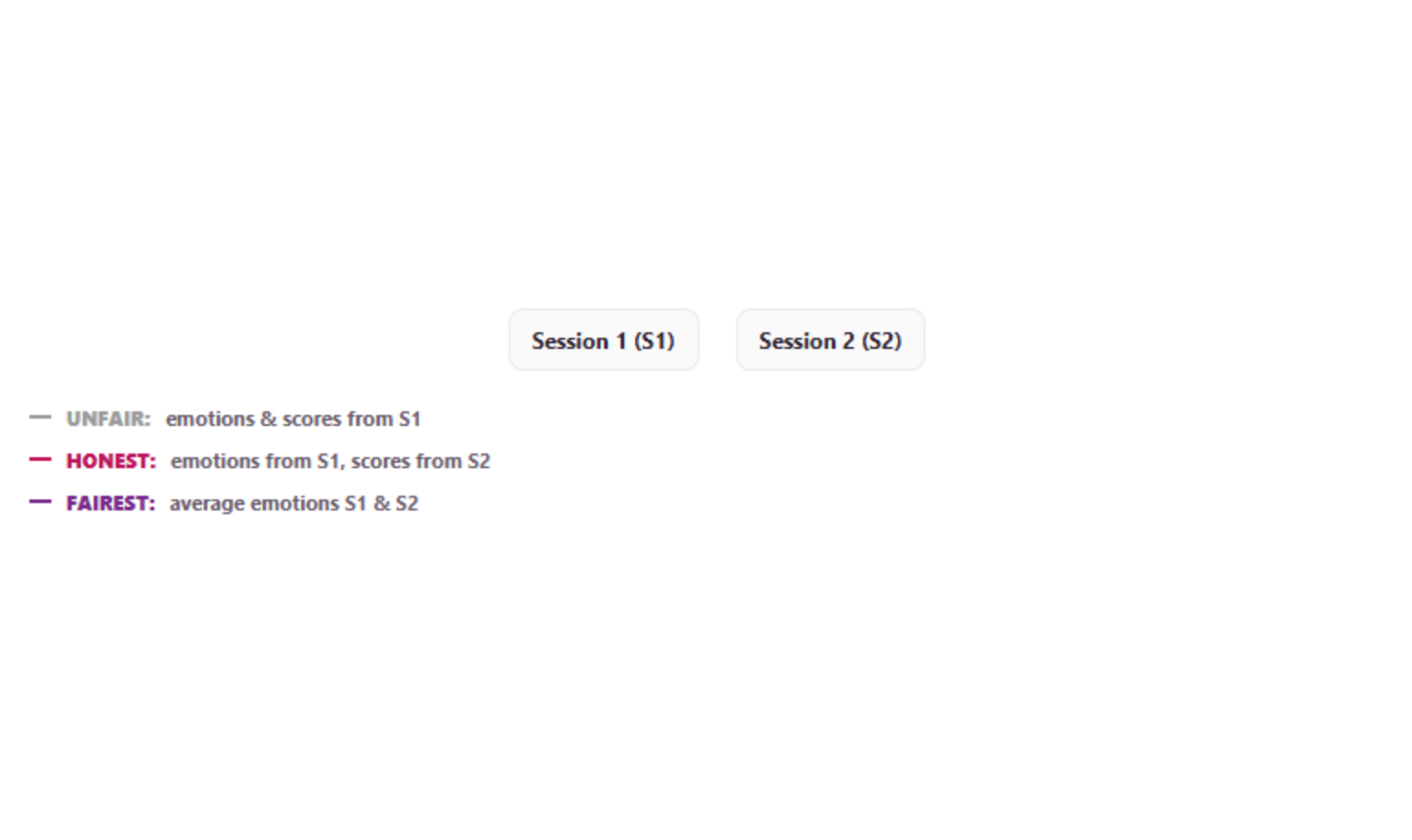


The answer depends on how we measure the emotions.

EMOTION SOURCES

Test-Retest Setup

The same user rated the exact same photo twice across distinct sessions (S1 & S2).



* only P changes; S and S-global stay fixed

Measure	Perception (CCC)	Winner
same-session	58% [52%, 65%]	feeling
cross-session	38% [30%, 46%]	weighting
averaged (fairest)	55% [48%, 62%]	≈ 50/50 (tie) both matter

● can't tell → both perception & weighting are equally important.

so we report it honestly, as a supporting result with ranges

The realistic ceiling is 0.64.

0.639

Deployment-Realistic Ceiling

Aesthetic predictability ceiling measured across distinct sessions (Cross-Session).

PERFORMANCE RATIO

59%

we reach 59% of it (0.380 / 0.639)



and averaging emotions raises it — the limit is measurement noise



Human Noise Ceiling Estimates

Comparing different noise ceiling definitions with Hybrid model performance

Same-Session (Theoretical): **Higher (~0.8)**

Cross-Session (Deployment-Realistic): **0.639**

Our Hybrid Model: **0.380 (59%)**

Emotion makes personalizing worth it

THE 50-RATINGS THRESHOLD

When is Personalization Worthwhile?

Below approximately 50 user ratings, individual personalization performs worse than the population average model due to data sparsity.

CRITICAL PRACTICAL INSIGHT

“Unless you use emotion mediation (Hybrid), personalizing with few ratings (under 50) is worse than simply using the population average.”

The Hybrid model quickly outperforms the population average past 50 ratings, whereas the Direct baseline fails to beat the population average even with more data.

Below 50 ratings, personalizing hurts. Direct never beats the crowd; only Hybrid does, from 50 on.



Cold-Start Threshold Curve

Hybrid personalization begins to strictly outperform population baseline after 50 user ratings

Population (dashed pink): **threshold limit**

Hybrid (purple): **beats crowd at 50**

Direct (grey): **fails to beat crowd**

Validation: Placebo Control is Random Noise.

CONTROL EXPERIMENT RESULTS (CCC)

Ours (Real Emotions)	0.380
Placebo (Random Noise / PCA)	0.160

Ours: 0.380, Placebo: 0.160. Emotion semantics are real.

MECHANISM VERIFICATION

Three tests — do they point to “ceiling” or “different mechanism”?

1. Partial correlation — control for Direct strength

leans ceiling

If it's a ceiling, art's correlation should recover once baseline strength is removed.

art: +0.13 (*ns*) → +0.18 · $p = 0.048$

2. Spread of the gain within each category (SD)

weak / borderline

If art is capped, its gain should vary the least — lowest spread.

art SD = 0.054 (lowest) · Levene $p = 0.21$

3. Interaction test — is the *emo_r*→gain slope different in art?

no evidence for H2

If art had a different mechanism, this would show up clearly.

$p = 0.056$ (not significant)

All three lean toward “ceiling” — but every number sits on the 0.05 border

So: no evidence art works differently — not proof they're the same

Key takeaways.

1 Explainable Emotion Model
Built an explainable model that predicts preference through 7 intermediate emotions.

2 Trait-Free Parity
Comparable to trait-based baselines while using only emotion (protecting user privacy).

3 Accuracy-Driven Gain
Helps more when we predict a person's emotions more accurately (\$emo_r\$).

4 Ceiling & Cold-Start
Realistic ceiling is 0.63, and emotion is what makes personalizing worthwhile (from 50 ratings).

“Emotion is a powerful, explainable bridge to subjective taste.”

Future work.

- **Context-Dynamic Weighting**

Weights dynamically update depending on context, ambient conditions, or user mood shifts over time.

- **Active Learning Querying**

Query users strategically to fit weights faster, reducing the 50-ratings cold-start threshold significantly.

Thank you

QUESTIONS & ANSWERS

Table 1: Comprehensive Model Metrics.

Q&A-1 ALL PERFORMANCE METRICS

Model Type	Traits Required	CCC	PLCC	SRCC
Direct Baseline (VLM Only)	No	0.293	0.301	0.297
ICI Baseline (8B VLM)	Yes (Big Five Traits)	0.369	0.372	0.365
MIR Baseline (8B VLM)	Yes (Big Five Traits)	0.385	0.390	0.382
Ours (Hybrid, Explainable)	None (Traits-Free)	0.380	0.386	0.378

Table 2: Dataset Domain Breakdown.

Q&A-2 XPASS-VIS DATASET BREAKDOWN

Visual Domain	Evaluators	Unique Images	Total Ratings
Artistic Images	129	2,186	29,278
Fashion Photos	129	2,170	29,270
Landscape Photos	129	2,170	29,288
Total Dataset (XPASS-Vis)	129	6,526	87,836

Placebo: PCA Details & Formula.

Q&A-4 PCA BASELINE CONSTRUCTION

How are the placebo PCA features projected?

To check if the gain is from actual emotional semantics, we swap emotion dimensions with PCA features of the same length (7 dimensions) extracted from VLM features:

$$z^{PCA}(i) = f^{VLM}(i) \cdot W^{PCA} \text{ (7-dim PCA projection)}$$

$$Score(u,i) = w^0 + w^{u,1} z^{PCA,1}(i) + \dots + w^{u,7} z^{PCA,7}(i)$$

★ **Placebo Result:** The PCA baseline drops personalization CCC gain from **0.380** down to **0.160**. This verifies that actual human emotional semantics drive aesthetic personalization.

Analysis: Partial Correlation Driver.

Q&A-5 PARTIAL CORRELATION ANALYSIS

Controlling for the Direct baseline model's performance confirms that emotion predictability (*emo_p*) is the true driver of gain:

$$r_{gain, emo_r \cdot Direct} = 0.42 \quad (p < 0.001)$$

Key Insight: Emotion accuracy (*emo_p*) is the primary driver of accuracy gain, independent of how much prediction room the baseline provides.

Design Choice: Why Linear & Why 7 Emotions?

Q&A-7 MODEL ARCHITECTURE RATIONALE

WHY LINEAR MODEL?

- **Cognitive Science:** Linear weighting on mediated emotions matches human cognitive findings (Iigaya et al. 2020).
- **Explainability:** Yields a transparent per-person weight vector ($w_{u,c}$).
- **Regularization:** Ridge regression handles emotion collinearity without overfitting.

WHY EXACTLY 7 EMOTIONS?

- **Taxonomy:** Selected from the 21 emotional sub-scales in AESTHEMOS.
- **No Circularity:** Excludes adjectives like “beautiful” to prevent predicting preference from preference.
- **Sparsity:** Small feature space prevents overfitting while preserving granularity.

Related Work: Comparison with Prior PIAA.

Q&A-8 COMPARISON WITH PRIOR PIAA LITERATURE

Prior Work	Approach	Our Key Difference
Ryu & Yanaka (2022)	Big Five personality traits to group users.	Traits-free: Bypasses intrusive surveys using emotion semantics.
Liu & Wagemans (2020)	Shared (generic) consensus emotional ratings.	Personal weights: Adapts weighting profiles per user.
Lan et al. (2021)	Deep end-to-end black-box personalization.	White-box explainability: Yields readable user weights.
Iigaya et al. (2020)	Linear weighting on low-level features.	High-level VLM features: Models semantic visual emotions.